Lambda with API GATEWAY

0) What you are building

```
Browser → API Gateway → Lambda → JSON response
```

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1) Create Lambda

Go to AWS Lambda

Click

```
Create function
```

Fill details

Field	Value
Author from scratch	✅
Function name	hello-api
Runtime	Node.js 18.x
Architecture	x86_64

👉 Click **Create function**

---

2) Add code

You will see file:

```
index.mjs
```

Replace code with:

```
export const handler = async (event) => {
  return {
    statusCode: 200,
    headers: {
      "Content-Type": "application/json"
    },
    body: JSON.stringify({
      message: "Hello from Lambda 🚀",
      method: event.requestContext?.http?.method,
      path: event.rawPath,
      query: event.queryStringParameters || {}
    })
  };
};
```

Click

```
Deploy
```

3) Test Lambda (important)

Click:

```
Test → Create test event → Test
```

👉 You should see:

```
{
  "statusCode": 200
}
```

4) Create API

Go to Amazon API Gateway

Go to Amazon API Gateway

Click

```
Create API → HTTP API → Build
```

---

5) Add Integration

Click:

```
Add integration → Lambda
```

Select:

```
hello-api
```

Click **Add**

---

6) Add Route

Click:

```
Add route
```

Enter:

```
GET /hello
```

👉 Attach Lambda integration

Save

---

7) Create / Deploy

Click:

```
Next → Deploy
```

Stage

Select:

```
$default  (or create "dev")
```

---

8) Get endpoint

You will see:

```
https://xxxxx.execute-api.region.amazonaws.com
```

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9) Final URL

```
https://xxxxx.execute-api.region.amazonaws.com/hello
```

OR (if dev stage):

```
https://xxxxx.execute-api.region.amazonaws.com/dev/hello
```

---

10) Test in browser

Open URL → should return:

```
{
  "message": "Hello from Lambda 🚀"
}
```

---

11) Enable CORS (important for frontend)

Go to API Gateway:

```
CORS → Enable
```

Set:

```
Access-Control-Allow-Origin: *
```

Save

12) Done

🔥 What each step means

Step	Purpose
Lambda	runs your code
API Gateway	gives public URL
Route	maps URL → Lambda
Deploy	makes API live

⚠️ Common mistakes

Problem	Fix
500 error	wrong code format
404	route missing
CORS error	enable CORS
Not working	forgot deploy

🧠 Final understanding

```
Lambda = function
API Gateway = URL
Route = mapping
```

🚀 Next things you should try

1. POST API
2. Connect your S3 frontend
3. Save data (DB)
4. Add authentication

If you want, I can next:

- Build full CRUD API
- Connect your frontend project
- Add login system

Just tell me 👍